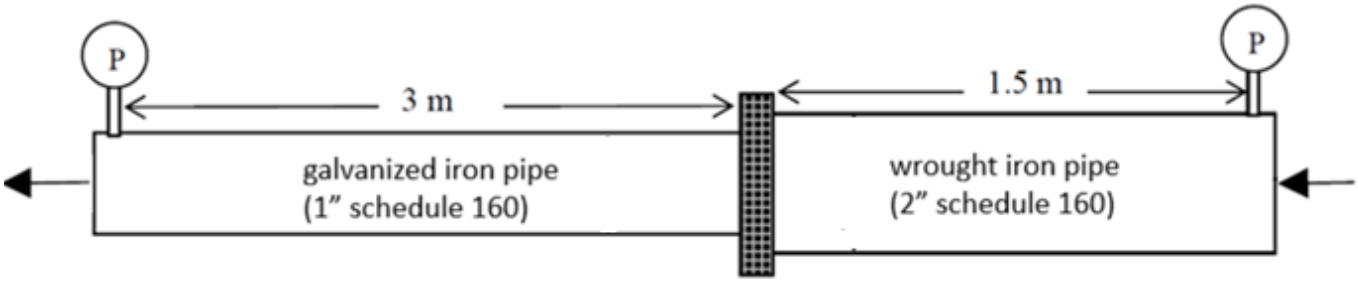
ChE 2O04 Tutorial #9

Tutorial Problem #1

Carbon tetrachloride at ambient temperature (viscosity = 0.001 Pa·s) is flowing through the horizontal piping system as shown below (not drawn to scale). A contraction fitting (also known as a reducing bushing) is used to connect the larger wrought iron pipe (2" schedule 160) to the smaller galvanized iron pipe (1" schedule 160).



a. Starting from a steady-state mass balance, write the appropriate equation that relates the average velocities in the two pipes.

b. Write the simplified form of the Bernoulli equation for the condition where reference points 1 & 2 are chosen to be at the centreline of the piping system where the two pressure gauges are located.

c. If the upstream pressure gauge reads 192 kPa and the downstream pressure gauge reads 20 kPa, determine the volumetric flow rate of carbon tetrachloride (m^3/s) in the piping system.